

# University of Dhaka



## Assignment - 3

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**Course Title: Advanced Econometrics-II**  
**Course Code: 408**

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### Question:

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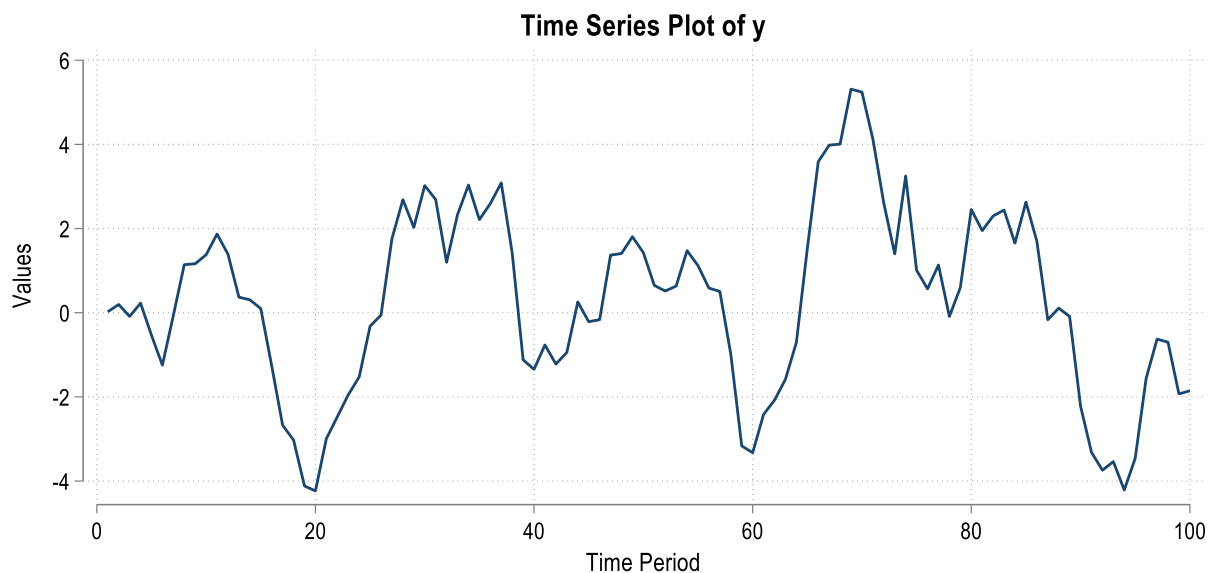
Attached is a STATA data file with a time series variable "y". First, using the Box-Jenkins methodology, determine which ARMA model this variable follows. After estimating the ARMA model, please forecast 10 steps ahead of "y" with a 95% confidence interval. Extra marks for better visual representation.

### Solution:

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Following the Box-Jenkins Methodology, we will follow (a) Identification (b) Estimation (c) Diagnostic Test, these 3 procedures to find the best Univariate ARMA model that this variable y follows.

#### *(a) Identification:*

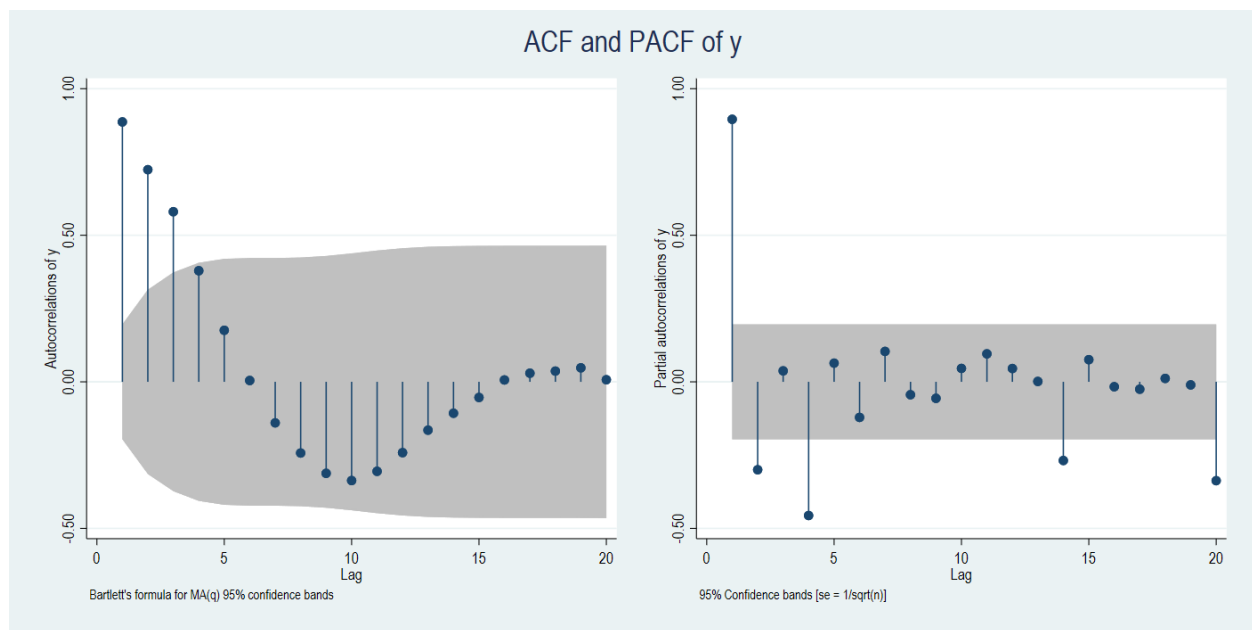


H0: Random walk without drift,  $d = 0$

|      | Test       | Dickey Fuller<br>Critical Values |        |        |
|------|------------|----------------------------------|--------|--------|
|      | Statistics | 1%                               | 5%     | 10%    |
| Z(t) | -3.622     | -3.517                           | -2.894 | -2.582 |

MacKinnon approximate p-value for Z(t) = 0.0054

The Time Series plot of the variable y clearly shows no significant trend. And the Augmented Dickey-Fuller Test also rejects the Null Hypothesis of a Unit Root Process at lag 4. So, we can conclude that variable y is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of y variable show significant spikes till lag 3 and exponential decay. So, I assume the presence of ARMA (2,3), ARMA (3,2), ARMA (3,3), ARMA (3,4), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

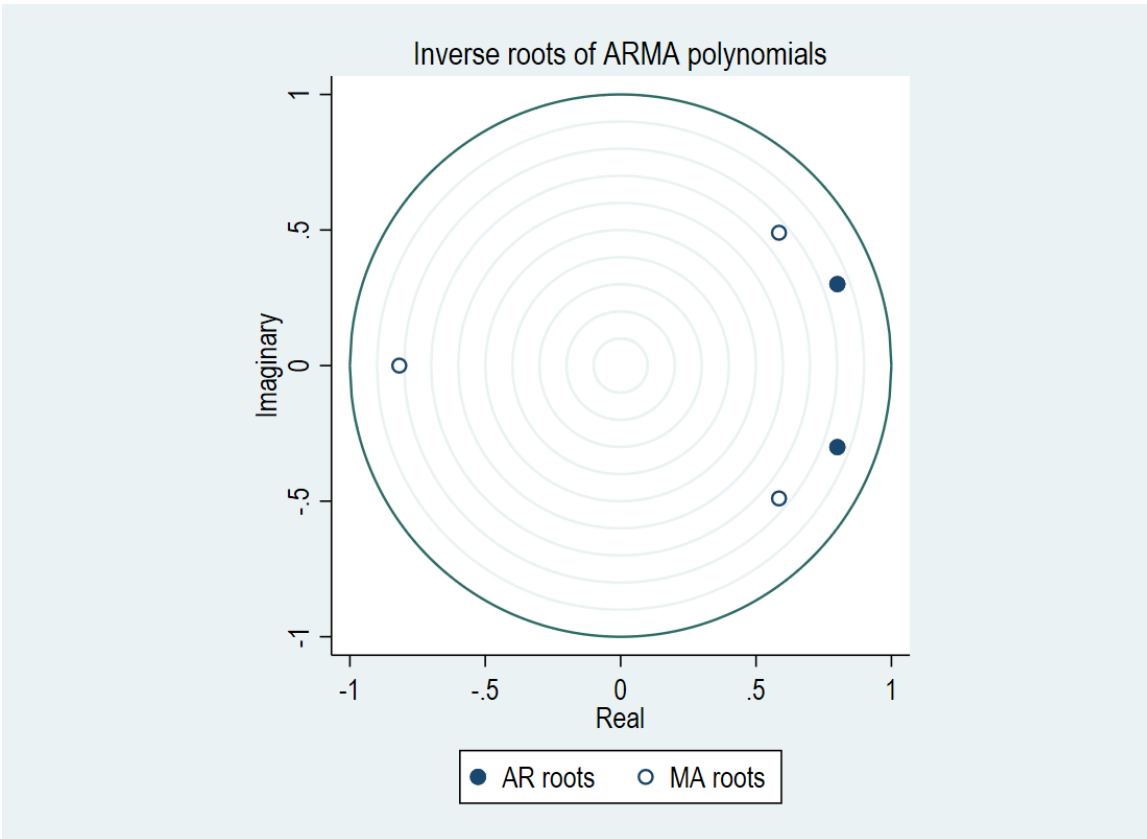
| <i>Model</i> | <i>N</i> | <i>LL (model)</i> | <i>df</i> | <i>AIC</i> | <i>BIC</i> |
|--------------|----------|-------------------|-----------|------------|------------|
| ARMA (2,3)   | 100      | -124.3935         | 7         | 262.7871   | 281.0233   |
| ARMA (3,2)   | 100      | -125.8966         | 7         | 265.7933   | 284.0295   |
| ARMA (3,3)   | 100      | -124.1635         | 8         | 264.327    | 285.1684   |
| ARMA (3,4)   | 100      | -123.6252         | 9         | 265.2504   | 288.6969   |
| ARMA (4,3)   | 100      | -123.5017         | 9         | 265.0035   | 288.45     |

Among those guessed model, ARMA (2,3) yields the lowest AIC and BIC so this is my chosen Univariate model for the variable y.

*(b) Estimation:*

| <b>ARIMA (2,3) regression</b> |       |         |                    |         |           |           |     |
|-------------------------------|-------|---------|--------------------|---------|-----------|-----------|-----|
| y                             | Coef. | St.Err. | t-value            | p-value | [95% Conf | Interval] | Sig |
| Constant                      | .219  | .51     | 0.43               | .668    | -.781     | 1.218     |     |
| ar                            |       |         |                    |         |           |           |     |
| L                             | 1.601 | .146    | 10.94              | 0       | 1.314     | 1.888     | *** |
| L2                            | -.731 | .12     | -6.12              | 0       | -.966     | -.497     | *** |
| ma                            |       |         |                    |         |           |           |     |
| L                             | -.352 | .145    | -2.43              | .015    | -.636     | -.068     | **  |
| L2                            | -.374 | .127    | -2.95              | .003    | -.623     | -.126     | *** |
| L3                            | .476  | .116    | 4.11               | 0       | .249      | .702      | *** |
| Constant                      | .828  | .057    | 14.52              | 0       | .716      | .94       | *** |
| Mean dependent var            |       | 0.275   | SD dependent var   |         |           | 2.162     |     |
| Number of obs                 |       | 100     | Chi-square         |         |           | 867.399   |     |
| Prob > chi2                   |       | .       | Akaike crit. (AIC) |         |           | 262.787   |     |

\*\*\*  $p < .01$ , \*\*  $p < .05$ , \*  $p < .1$



After estimating the ARMA (2,3) model we can see all the lag coefficients of AR and MA process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR and MA process are within the unit circle. So, we can say ARMA (2,3) model is acceptable for estimation.

(c) Diagnostic Test:

Portmanteau test for white noise

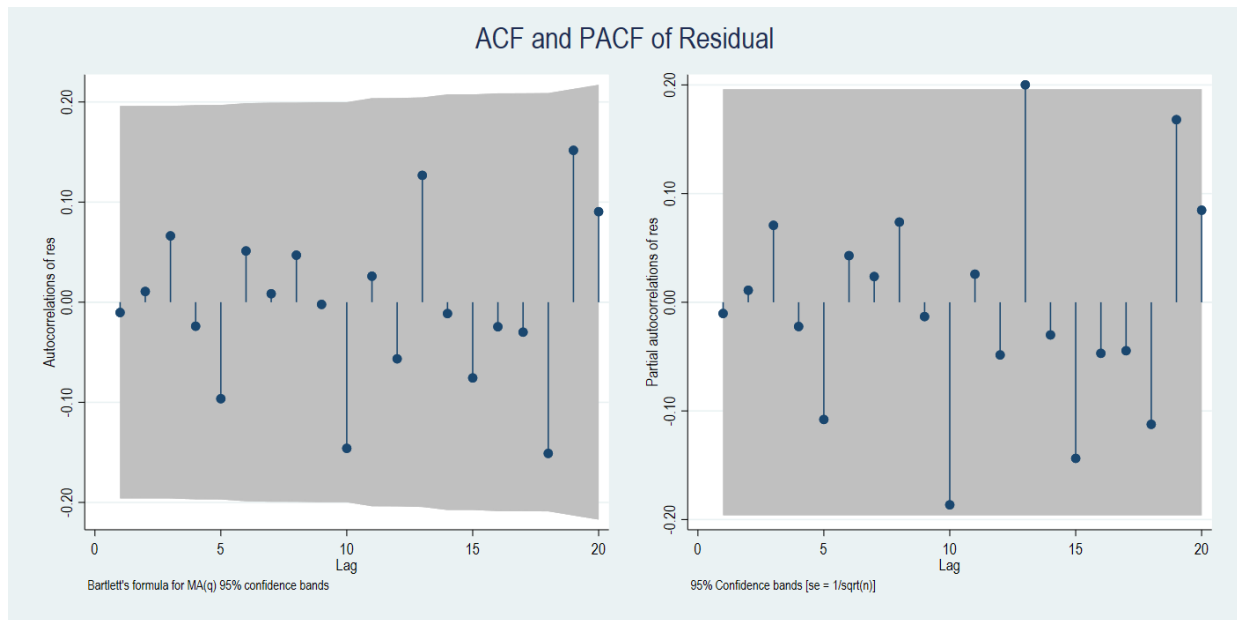
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Portmanteau (Q) statistics = 33.0943

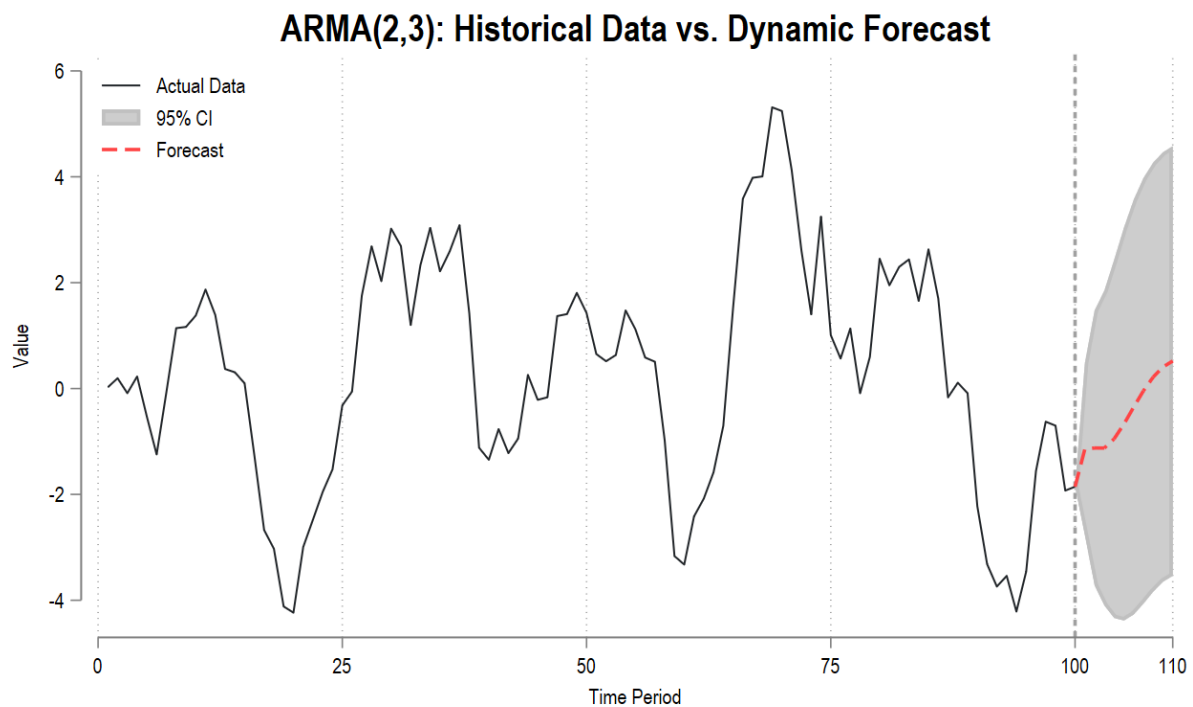
Prob > chi2(40) = 0.7721

The Ljung-Box test or Q test at lag 40 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 20. This diagnostic test satisfies the residuals being

white noise and according to the steps of Box Jenkins Methodology ARMA (2,3) model is accepted as the univariate process of the variable  $y$ .



*(d) 10 Steps Ahead Forecast:*



```

1  cd "F:\ECON 408\Practice do file"
2
3
4  clear all
5  set seed 1000
6
7  use "HOMEWORK 3.dta" , clear
8
9  rename t time
10 tsset time
11
12
13  *                               Identification
14  ****
15
16  *(1) Time Series plot of variable y
17  //-----
18
19
20  set scheme white_tableau
21
22  tsline y, ///
23      lcolor(navy) lwidth(medthick) ///
24      title("{bf:Time Series Plot of y}", size(large)) ///
25      xtitle("Time Period", size(medlarge)) ///
26      ytitle("Values", size(medlarge)) ///
27      ylabel(, labsize(medium)) ///
28      tlabel(, labsize(medium)) ///
29      graphregion(color(white)) xsize(8) ysize(4) ///
30      name(tsplot_hw3, replace)
31
32  graph export tsplot_hw3.png, name(tsplot_hw3) replace
33
34
35  *(2) Stationarity Test
36  //-----
37
38
39  varsoc y , maxlag(12)

```

```

40 dfuller y, lags(4)
41
42
43
44 *(3) ACF and PACF
45 //-----
46
47
48 set scheme s2color
49
50 ac y, name(acf_plot,replace) lags(20)
51 pac y, name(pacf_plot, replace) lags(20)
52 graph combine acf_plot pacf_plot, name(ACF_and_PACF_hw3,
replace) ///
53     title("ACF and PACF of y") xsize(8) ysize(4)
54
55 graph export ACF_and_PACF_hw3.png, name(ACF_and_PACF_hw3)
replace
56
57 * Assumed models: ARMA(2,3) , ARMA(3,2), ARMA(3,3),
ARMA(3,4), ARMA(4,3)
58
59 qui arima y, ar(1/2) ma(1/3)
60 estat ic
61
62 qui arima y, ar(1/3) ma(1/2)
63 estat ic
64
65 qui arima y, ar(1/3) ma(1/3)
66 estat ic
67
68 qui arima y, ar(1/3) ma(1/4)
69 estat ic
70
71 qui arima y, ar(1/4) ma(1/3)
72 estat ic
73
74 * ARMA(2,3) has the lowest AIC and BIC
75

```

```

76
77
78      *                      Estimation
79      *****
80
81      arima y, ar(1/2) ma(1/3)
82      estat ic
83      estat aroots, name(arootshw3, replace)
84      graph export arootshw3.png, name(arootshw3) replace
85
86
87      *                      Diagnostic Test
88      *****
89
90
91      predict res , residuals
92
93      ac res, name(res_acf, replace) lags(20)
94      pac res , name(res_pacf, replace) lags(20)
95      graph combine res_acf res_pacf , xsize(8) ysize(4) title(
96      "ACF and PACF of Residual") ///
97      name(res_acf_pacf_hw3, replace)
98      graph export res_acf_pacf_hw3.png, name(res_acf_pacf_hw3)
99      replace
100
101      wntestq res
102
103      *                      10 Days Ahead Forecast
104      *****
105
106
107      keep if time <= 100
108      qui arima y, ar(1/2) ma(1/3)
109      tsappend, add(10)
110
111      capture drop y_forecast y_mse y_se ci_lower ci_upper
112      predict y_forecast, y dynamic(101)

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```

113 predict y_mse, mse dynamic(101)
114
115 gen y_se = sqrt(y_mse)
116 gen ci_lower = y_forecast - 1.96 * y_se
117 gen ci_upper = y_forecast + 1.96 * y_se
118
119 qui replace y_forecast = y if time == 100
120 qui replace ci_lower = y if time == 100
121 qui replace ci_upper = y if time == 100
122
123 list time y_forecast ci_lower ci_upper if time >= 99 & time
    <= 103
124
125
126 set scheme white_tableau
127
128 twoway (rarea ci_lower ci_upper time if time >= 100, color(
    "gs12") lwidth(medthick)) ///
129       (line y time if time <= 100 & time >= 1, lcolor("33
    37 41") lwidth(medthin)) ///
130       (line y_forecast time if time >= 100, lcolor("255 68
    69") lpattern(dash) lwidth(medthick)), ///
131       xline(100, lcolor(gs10) lwidth(medthick) lpattern(
    shortdash)) ///
132       graphregion(color(white)) ///
133       title("{bf:ARMA(2,3): Historical Data vs. Dynamic
    Forecast}") ///
134       xtitle("Time Period") ///
135       ytitle("Value") ///
136       legend(order(2 "Actual Data" 1 "95% CI" 3 "Forecast")
    pos(11) ring(0) ///
137       cols(1)) ///
138       ylabel(, format(%9.0f) nogrid) ///
139       xlabel(0(25)100 110, labsize(small)) ///
140       xsize(6.5) ysize(4) name(forecast_plot, replace)
141
142 graph export forecast_plot.png, name(forecast_plot) replace
143
144

```